

ALLEN SCHULTZ

Linux Systems Administrator | DevOps Engineer | Network Infrastructure

Medford, OR 97504 | 801-550-0865 | allen.schultz@gmail.com

LinkedIn: [linkedin.com/in/allen-schultz-89810014](https://www.linkedin.com/in/allen-schultz-89810014) | GitHub: github.com/PhoenixCullen

PROFESSIONAL SUMMARY

Linux Systems Administrator with 8+ years of technical experience spanning enterprise Linux administration, network infrastructure, security, and automation. Proven expertise in Debian-based systems, virtualization, infrastructure automation, and secure system design. Currently pursuing Linux Foundation certifications (DevOps/Security track) and full-stack development training (Boot.dev - Python/Go) to expand infrastructure-as-code and automation capabilities. Active Qubes OS security research on System76 hardware demonstrating advanced virtualization and security architecture skills. Seeking remote Linux Administrator, DevOps Engineer, or Network Infrastructure role leveraging deep technical skills and continuous learning commitment.

Core Competencies:

Linux Administration (Debian, Ubuntu, Arch) • DevOps Practices & Automation • Network Infrastructure & SDN • Security Architecture • Bash Scripting & Python • VoIP (FreeSWITCH, SIP) • Virtualization (Qubes OS, VirtualBox, KVM) • Docker & Containerization • Git/GitLab Version Control • Infrastructure Security (LUKS, VM Isolation) • PXE/TFTP Boot Environments

TECHNICAL SKILLS

Operating Systems: Linux (Debian, Ubuntu, Arch, Manjaro, Qubes OS), Windows Server 2003-2022, Windows 11

Virtualization: Qubes OS (Xen hypervisor), VirtualBox, KVM, GPU passthrough, VM security isolation

Storage & Filesystems: Btrfs (subvolumes, snapshots, compression), LVM, LUKS encryption, RAID configurations

Programming & Scripting: Bash, Python (in training), Go (in training), C/C++ (academic), basic Perl/PHP

Networking: Cisco IOS, routing/switching, VLANs, network segmentation, Software Defined Networking (SDN) concepts

VoIP: FreeSWITCH PBX, SIP phone provisioning and automation

DevOps & Automation: Docker, Git/GitLab, automated deployment scripts, infrastructure automation

Security: LUKS full-disk encryption, VM-based security isolation, CompTIA Security+, secure architecture design

Tools: PXE/TFTP boot servers, package management (apt, pacman, yum), Timeshift backup solutions, systemd

TECHNICAL PROJECTS & LAB EXPERIENCE

Qubes OS Security Research Platform (System76 Bonobo WS) - 2024-Present

High-performance security-focused virtualization and learning environment

Hardware Platform:

- Intel Core i9-14900HX (24 cores, 32 threads), 64GB DDR5 RAM
- Dual-GPU architecture: Intel Raptor Lake-S UHD Graphics (dom0) + NVIDIA GeForce RTX 4080 Mobile (passthrough-capable)
- 3x 3.6TB NVMe drives (10.8TB total enterprise-grade storage)
- System76 Bonobo WS professional Linux workstation

Current Implementation:

- Running Qubes OS 4.3 RC3 (pre-release Xen-based security hypervisor)
- Implemented security-by-isolation architecture with compartmentalized VMs:
 - Vault qube (air-gapped) for cryptographic key storage
 - Separate qubes for personal, work, and untrusted activities
 - Whonix integration for anonymized networking research
- LUKS-encrypted storage with hardware-backed VM isolation (Intel VT-x, VT-d)
- Evaluating GPU passthrough architecture for isolated high-performance workloads

Skills Demonstrated:

- Xen hypervisor administration and VM security isolation
- Security-by-compartmentalization model implementation
- Pre-release software testing and hardware compatibility troubleshooting
- GPU passthrough concepts and dual-GPU architecture
- System76 open-hardware platform and Linux-first design
- Whonix/Tor integration for privacy research

Why This Platform:

- System76 hardware chosen for Linux-first design and open-source firmware support

- Running RC3 pre-release to stay current with cutting-edge security technologies
 - Portable lab environment demonstrating enterprise security concepts
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Advanced Storage Architecture Design (Desktop Workstation - In Planning)

Enterprise-grade storage solution design for production deployment

Target Hardware (AMD Desktop):

- AMD Ryzen 9 7950X3D (16 cores), 64GB RAM, AMD Radeon RX 7800 XT
- 5-drive array totaling 15TB+:
 - nvme1n1: 1.9TB NVMe (performance-critical system storage)
 - sda: 10.9TB (home directory with isolated Steam games subvolume)
 - sdb: 2.7TB, sdc: 931.5GB, nvme0n1: 465.8GB (aggregated data storage)

Planned Architecture:

- Btrfs on LVM on LUKS across all drives for security and flexibility
- Three-tier volume group strategy:
 - **vg-system** (nvme1n1): root, var/log, var/cache, swap, docker
 - **vg-data** (sdb+sdc+nvme0n1): VMs, aggregated capacity storage
 - **vg-home-allen** (sda): home directory + separate Steam games subvolume
- Subvolume isolation with independent compression policies:
 - Root and home: zstd compression with Timeshift snapshot management
 - var/log and var/cache: Separate filesystems, no snapshots (performance)
 - Steam games: Dedicated subvolume with independent snapshot schedule
- Full LUKS encryption across all volume groups (5 encrypted volumes)
- Automated disaster recovery with Timeshift integration

Skills to Demonstrate:

- Complex LVM topology design across heterogeneous storage
- Btrfs advanced features: subvolumes, snapshots, compression, send/receive
- Multi-tier storage strategy for different workload performance requirements

- Security-first design with layered encryption
- Backup and disaster recovery architecture planning
- Storage migration planning with zero data loss

Status: Complete architectural design documented, awaiting implementation window

Multi-System Administration Portfolio (Ongoing)

Active management of diverse Linux environments

Current Infrastructure:

- Primary desktop workstation (Arch-based custom build)
- System76 Bonobo WS laptop (Qubes OS 4.3 RC3)
- Friend assistance: Windows 11 to Linux migration, spyware removal, security hardening
- Total: 3-5 systems under active management

Activities:

- Mixed Linux/Windows VM environments for testing and compatibility
- Git/GitLab collaboration workflows (contributing to MegaMek/MekHQ open-source project)
- Real-world user support and migration assistance
- Cross-platform troubleshooting and system administration

Professional Development in Progress:

- Boot.dev Full-Stack Development Program: Python and Go backend development, infrastructure automation
 - Linux Foundation Certification Track: Focusing on LFCS (System Administrator) with DevOps and Security specializations
 - Software Defined Networking (SDN): Self-study of modern network virtualization and automation concepts
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PROFESSIONAL EXPERIENCE

Harry and David, Medford, OR

Seasonal Data Entry Associate | September 2023–February 2025

Seasonal employment during peak holiday periods (Sep-Feb annually)

- Processed 20+ orders daily into computer systems with high accuracy during peak holiday volume periods
- Verified order data integrity and reconciled discrepancies in time-sensitive e-commerce environment
- Completed three seasonal deployments: 2023-24, 2024-25, and 2025-26 seasons
- Maintained consistent performance metrics across multiple seasonal deployments in fast-paced environment

Superior Service Transport, Salt Lake City, UT

Driver Check-in Clerk/Scheduling | April 2023–July 2023 (Contract)

- Managed logistics coordination for transportation dispatch and delivery scheduling
- Processed truck trip manifests, verified documentation accuracy, maintained delivery database
- Coordinated with drivers and endpoint facilities ensuring on-time delivery execution
- Handled high-volume documentation processing in time-critical logistics environment

RidgeCrest Herbals, Salt Lake City, UT

Linux System Administrator & VoIP Administrator | May 2012–February 2014

Primary Linux infrastructure and VoIP services administrator for herbal supplement manufacturer

Linux Infrastructure Management:

- Administered heterogeneous Linux environment supporting 15-person team across Debian, Ubuntu, and Linux Mint distributions
- Managed server infrastructure: file servers, application servers, development environments, backup systems
- Implemented automated backup solutions using bash scripting and Git-based configuration management
- Administered VirtualBox virtualization platform for development, testing, and legacy system support
- Maintained multiple access methods for critical legacy DOS database: VirtualBox VMs and DOSBox emulation, ensuring business continuity for aging but essential systems

- Reduced system downtime 40% through proactive monitoring scripts and automated maintenance procedures
- Maintained Windows XP/7 client systems alongside Linux infrastructure for mixed-OS environment

VoIP Infrastructure Migration & Automation:

- Migrated company telephony from legacy Asterisk PBX to FreeSWITCH, evaluating modern VoIP platforms for improved features and reliability
- Engineered FreeSWITCH-based PBX system from ground-up, replacing aging infrastructure with scalable open-source solution
- Created automated SIP phone provisioning templates deployable via PXE/TFTP boot environment
- Deployed 15 user SIP phones plus voicemail system, fax integration, printer, and 6 support lines with hunt group configuration
- Designed scalable system capable of supporting 40+ endpoints using standardized templates
- Reduced telephony costs 60% by migrating from traditional PBX to open-source VoIP solution
- Integrated VoIP system with existing network infrastructure maintaining QoS and call quality standards

Technical Achievements:

- Designed automated backup system with Git integration for configuration versioning and change tracking
- Established standardized Linux workstation deployment process reducing setup time from 4 hours to 30 minutes
- Created and maintained bash script library for common administrative tasks (user management, backup automation, system updates)
- Maintained 99%+ uptime for critical business systems throughout tenure
- Documented all systems and procedures for knowledge transfer and disaster recovery

ITT Systems (DoD Contractor), Afghanistan

Network Administrator | September 2009–May 2010

Security-cleared network administrator for Department of Defense communications infrastructure in combat zone

- Maintained dual-tier classified network infrastructure in hostile environment supporting 200+ users
- Administered Cisco switching and routing infrastructure: VLANs, ACLs, routing protocols for secure network segmentation

- Managed Microsoft Active Directory, Exchange Server 2003, and Windows Server 2003 domain services
- Implemented critical encryption protocol updates maintaining DoD security compliance standards
- Provided Tier 2/3 escalation support for SATCOM administrators and systems administrators, including assistance with Active Directory operations, password resets, account management, and Windows Client/Server updates
- Executed emergency network maintenance during critical operational periods
- Earned CompTIA Security+ certification during deployment (required for position)
- Maintained operational security (OpSec) protocols for classified communications infrastructure

Teleperformance USA, Salt Lake City, UT

Hardware Support Technician - Dell Tier 2 Contract | July 2007–October 2008

Advanced technical support specialist for Dell enterprise and consumer hardware platforms

- Achieved Dell Tier 2 Certification for advanced hardware diagnostics and complex troubleshooting
- Resolved 40+ technical issues daily via phone and internet chat with 85%+ customer satisfaction rating
- Served as Resolution Specialist: mentored junior agents on complex troubleshooting scenarios and escalation procedures
- Dispatched field technicians and coordinated parts replacement for critical system failures
- Diagnosed hardware failures across Dell product lines: servers, workstations, laptops, peripherals
- Contributed to successful launch of Core Business Group Internet Chat support channel as first-wave team member
- Maintained extensive product knowledge across multiple Dell Lines of Business

Internet Support Specialist - Clearwire Contract | April 2006–July 2007

- Provided Tier 2 technical support for wireless broadband services and emerging VoIP technologies
- Certified as VoIP Subject Matter Expert (SME): provisioned customers for VoIP services, troubleshoot SIP connectivity
- Earned Clearwire Email Specialist certification for advanced email system troubleshooting
- Assisted fellow agents as Tier 2 resource for complex technical escalations
- Troubleshoot wireless modem connectivity, email configuration, web hosting issues
- Documented known issues and outage information for customer communication

Convergys, Murray, UT

Provisional Technical Engineer - AT&T Managed Internet Services | August 2000–November 2001

Technical project manager for T1/Frame Relay business internet circuit installations

- Managed end-to-end installation projects for T1/FT1 business internet circuits: project planning through activation
- Configured customer premises equipment: Cisco and Paradyne routers, CSU/DSUs for circuit termination
- Coordinated with multiple AT&T vendors and third-party circuit providers for complex installations
- Managed IP address allocation, subnet design, and routing configuration for customer networks
- Troubleshoot layer 1-3 connectivity issues during circuit installation and activation phases
- Utilized network management tools for circuit testing and performance verification
- Maintained project documentation and customer communication throughout installation lifecycle
- Certified as AT&T Provisional Technical Engineer

@Home Internet Support Technician - AT&T Contract | April 2000–August 2000

- Provided technical support for cable modem internet services: connectivity, email, browsing issues
 - Troubleshoot customer equipment and network configuration problems
 - Coordinated with network operations center for outage information and escalations
 - Maintained high customer satisfaction scores in fast-paced support environment
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EDUCATION

Salt Lake Community College | Associate of Science, Computer Science (2001-2006, 2016-2018)

Linux & Security Focused Coursework:

- Certified Ethical Hacker (CSIS 2360) - 2018
- Linux System Administration II (CSIS 2132) - 2017
- Linux System Administration I (CSIS 1132) - 2017
- Introduction to Linux (CSIS 1032) - 2016
- Database Management (CIS 2600), Software Engineering (CS 2750), Data Communications/Networking (CS 2400)
- Computer Architecture (CS 2620), Discrete Structures (CS 2310)

Cisco Networking & Programming Specialization:

- Cisco Advanced Routing & Switching (CNT 160), Project Based Learning (CNT 170)
- Cisco Networking Fundamentals (CNT 140), Router Technology (CNT 150)
- C Programming I (CIS 162)
- Computer Maintenance I: A+ Preparation (BPC 170)

Boot.dev | Full-Stack Backend Development Program (In Progress)

Python and Go backend development, DevOps automation, infrastructure-as-code

CERTIFICATIONS

Active Certifications:

- **Linux Essentials** – Linux Professional Institute (2023)
- **CompTIA Security+** (2010) - Network security, cryptography, security operations
- **CompTIA A+** (2007) - Hardware and software troubleshooting

In Progress - Target 2025:

- **Linux Foundation Certified System Administrator (LFCS)** - System administration fundamentals
- **Linux Foundation Certified Engineer (LFCE)** - Advanced Linux engineering (planned)
- Linux Foundation certifications focused on DevOps and Security specializations

Historical Certifications:

- Cisco Certified Network Administrator (CCNA) - Routing, switching, network fundamentals
 - Dell Tier 2 Hardware Certification - Advanced Dell hardware diagnostics
 - AT&T Provisional Technical Engineer - T1/circuit installation and troubleshooting
 - CIW Web Foundations Associate - Web technologies and standards
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PROFESSIONAL PROFILE

Career Context: Extensive technical background with deep expertise in Linux administration, network

infrastructure, and security. Career timeline reflects periods of family caregiving responsibilities (2014-2023) during which continued advancing technical skills through personal infrastructure projects and self-directed study. Recent Linux Essentials certification and pursuit of Linux Foundation professional certifications demonstrate commitment to formalizing extensive hands-on experience with industry-recognized credentials.

Current Technical Focus:

- Expanding DevOps skillset: containerization, infrastructure-as-code, CI/CD pipelines
- Security specialization: secure architecture design, encryption, VM-based isolation (active Qubes OS research)
- Backend development: Python and Go for automation and infrastructure tools
- Software Defined Networking: modern network virtualization and programmability

Work Preferences: Strong preference for remote work enabling deep focus and eliminating commute time. Proven ability to work independently, self-manage complex projects, and maintain productivity in remote environments. Comfortable with distributed team collaboration tools and asynchronous communication.

References available upon request

Open to remote work opportunities nationwide

Salary requirement: \$46,500+ annually (\$22.36/hr minimum)